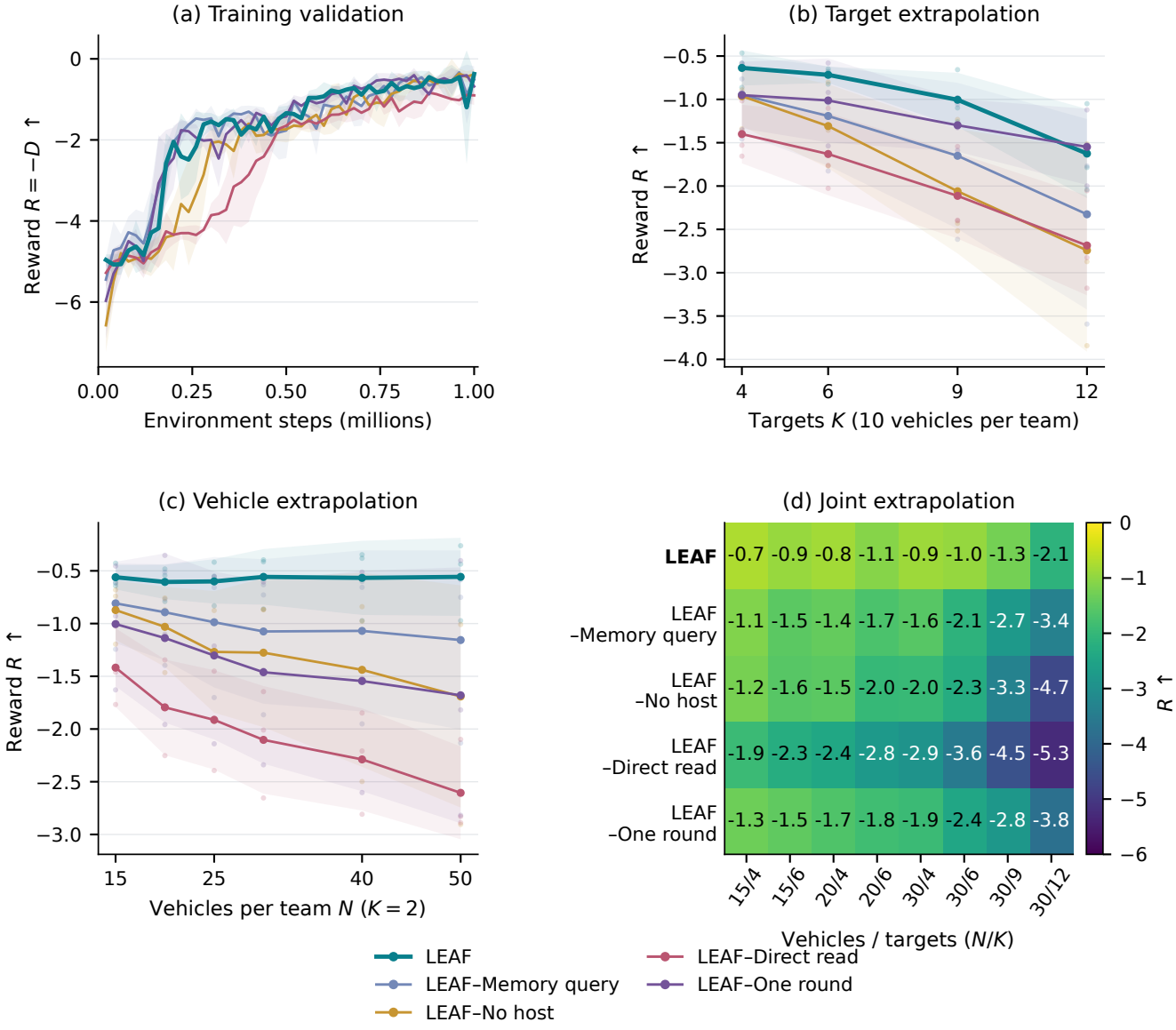

(a) Training validation
Reward $R = -D$ ↑
0
−2
−4
−6
0.00
0.25
0.50
0.75
1.00
Environment steps (millions)
(b) Target extrapolation
Reward R ↑
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
−3.5
−4.0
4
6
9
12
Targets K (10 vehicles per team)
(c) Vehicle extrapolation
Reward R ↑
−0.5
−1.0
−1.5
−2.0
−2.5
−3.0
15
25
40
50
Vehicles per team N ($K = 2$)
(d) Joint extrapolation
LEAF -0.7 -0.9 -0.8 -1.1 -0.9 -1.0 -1.3 -2.1
LEAF –Memory query -1.1 -1.5 -1.4 -1.7 -1.6 -2.1 -2.7 -3.4
LEAF –No host -1.2 -1.6 -1.5 -2.0 -2.0 -2.3 -3.3 -4.7
LEAF –Direct read -1.9 -2.3 -2.4 -2.8 -2.9 -3.6 -4.5 -5.3
LEAF –One round -1.3 -1.5 -1.7 -1.8 -1.9 -2.4 -2.8 -3.8
15/4
15/6
20/4
20/6
30/4
30/6
30/9
30/12
Vehicles / targets (N/K)
0
−1
−2
−3
−4
−5
−6
R ↑
LEAF
LEAF–Memory query
LEAF–No host
LEAF–Direct read
LEAF–One round